

Eric Elmoznino

Artificial Intelligence
Cognitive Neuroscience

Contact

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Websites



Education

PhD | Computer Science
Mila (Université de Montréal)
2022-2026 GPA: 4.3/4.3

MA | Cognitive Science
Johns Hopkins University
2019-2020 GPA: A+

BASc | Computer Engineering
University of Toronto
2014-2019 GPA: 3.95/4.0

DCS | Health Sciences
Dawson College
2012-2014 GPA: 96/100

Skills

Expertise

AI, Deep learning, Machine learning,
Computational neuroscience,
Psychology, Linguistics, Algorithmic
information theory

Spoken Languages

English, French (Fluent in both)

Interests

Mind, Brain, and AI

Consciousness, Compositionality,
Open-endedness, In-context learning,
Algorithmic information theory,
Causality, Knowledge acquisition, OoD
generalization, Lifelong learning

Public Speaking

Technical presentations, Teaching,
Scientific outreach

Other Disciplines

Data Science, Genetics, Astrophysics,
Philosophy of mind, Epistemology

Research positions & Work experience

- 2022-26 **AI Researcher (PhD)** Mila (Université de Montréal)
Inductive biases from conscious higher-level cognition ◦ Compression based perspectives on intelligence, compositionality, and open-endedness ◦ Supervised by Guillaume Lajoie & Yoshua Bengio
- 2024-26 **AI Student Researcher** Google – Paradigms of Intelligence, Remote
Open-ended exploration for intrinsic curiosity leveraging in-context learning ◦ Supervised by João Sacramento & Blaise Agüera y Arcas
- 2025, 26 **Visiting Researcher** Institute for Basic Science, South Korea
Brain-computer interfaces using fMRI & LLMs ◦ Models of consciousness, blindsight, and introspection ◦ Supervised by Hakwan Lau
- 2019-22 **Cognitive Science Researcher (MA)** Johns Hopkins University, Baltimore
Research on information representation and algorithms in the visual system of the human brain ◦ Supervised by Michael Bonner
- 2017-19 **Machine Learning Researcher** ModiFace, Toronto
Computer vision models for the beauty industry
- 2018 **Computer Vision Contractor** Precious, Remote
Computer vision models for automatic baby photo album generation
- 2016 **Software Developer Intern** Orbis Investments, Vancouver
Full-stack web development for internal workflow applications
- 2012-14 **Corporal, Infantry Division** Canadian Armed Forces (Reserves)
Participation in combat and reconnaissance exercises

Teaching experience

- 2020-25 **Instructor for data science bootcamp** Lighthouse Labs, Remote
Teach lectures on machine learning topics at a full-time data science bootcamp to students with no prior coding experience
- 2022 **TA for IFT3395/6390 (fundamentals of ML)** Université de Montréal
Lead groups of students through tutorial exercises ◦ Review lecture material and answer student questions
- 2021 **TA for computational neuroscience** Neuromatch Academy, Remote
Lead groups of students through tutorial exercises ◦ Review lecture material and answer student questions
- 2018 **Instructor for ECE1780 (AI on mobile device)** University of Toronto
Taught lectures for a graduate course on DNNs deployed to mobile devices under Prof. Parham Aarabi

Scholarships & Awards

- 2024 UNIQUE Conference Travel Award (\$1,000 value)
- 2023 Vanier Canadian Graduate Scholarship (\$150,000 value)
- 2023 Natural Sciences and Engineering Research Council of Canada (NSERC) PhD Grant (\$105,000 value)
- 2022 UNIQUE Neuro-AI Excellence Scholarship (\$15,000 value)
- 2016 Class of 4T3 Engineering James Ham Award (\$10,000 value)
- 2015 Class of 5T6 Award of Merit (\$15,000 value)
- 2013 First Choice Science Award
- 2012 McGill Science Award and Scholarship
- 2012 A.J. Grant Shield and Scholarship
- 2012 Quebec English Public Speaking (Provincial Finals) — Bronze Medal
- 2012 Governor General of Canada Academic Medal
- 2011 Quebec French Public Speaking (Provincial Finals) — Silver Medal

Publications

- 2026 **Can In-Context Learning Support Intrinsic Curiosity?** Eric Elmoznino, Sangnie Bhardwaj, Johannes von Oswald, Rajai Nasser, Blaise Agüera y Arcas, João Sacramento, Rif A Saurous, Guillaume Lajoie *NeurIPS oral*
- 2026 **Consciousness and artificial intelligence.** In *The Scientific Study of Consciousness: Experimental & Theoretical Approaches*, chapter 6.1. Eric Elmoznino, Jonathan Simon, Guillaume Lajoie, Yoshua Bengio *Forthcoming in Springer*
- 2026 **Simplifying the Modeling of Arbitrary Conditionals in Natural Language.** Yinhan Lu, Eric Elmoznino, Léo Gagnon, Sarthak Mittal, Tejas Kasetty, Guillaume Lajoie *Preprint*
- 2026 **A Compression Perspective on Simplicity Bias.** Tom Marty, Eric Elmoznino, Dhanya Sridhar *Preprint*
- 2025 **A Complexity-Based Theory of Compositionality.** Eric Elmoznino, Thomas Jiralerspong, Yoshua Bengio, Guillaume Lajoie *ICML poster*
- 2025 **In-context learning and Occam's razor.** Eric Elmoznino, Tom Marty, Tejas Kasetty, Léo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie *ICML poster*
- 2025 **Does learning the right latent variables necessarily improve in-context learning?** Eric Elmoznino, Sarthak Mittal, Léo Gagnon, Sangnie Bhardwaj, Tom Marty, Dhanya Sridhar, Guillaume Lajoie *ICML poster*
- 2025 **Illusions of AI consciousness.** Yoshua Bengio & Eric Elmoznino *Science*
- 2025 **Multi-agent cooperation through learning-aware policy gradients.** Alexander Meulemans, Seijin Kobayashi, Johannes von Oswald, Nino Scherrer, Eric Elmoznino, Blake Richards, Guillaume Lajoie, Blaise Agüera y Arcas, João Sacramento *ICLR poster*
- 2025 **Convolutional architectures are cortex-aligned de novo** Atlas Kazemian, Eric Elmoznino, Michael F. Bonner *Nature Machine Intelligence*
- 2024 **Amortizing intractable inference in large language models.** Edward J. Hu, Moksh Jain, Eric Elmoznino, Younesse Kaddar, Guillaume Lajoie, Yoshua Bengio, Nikolay Malkin *ICLR oral – best paper honorable mention*
- 2024 **Sources of Richness and Ineffability for Phenomenally Conscious States.** Eric Elmoznino, Xu Ji, George Deane, Axel Constant, Guillaume Dumas, Guillaume Lajoie, Jonathan Simon, Yoshua Bengio *Neuroscience of Consciousness*
- 2024 **High-performing neural network models of visual cortex benefit from high latent dimensionality.** Eric Elmoznino & Michael F. Bonner *PLOS Computational Biology*
- 2023 **Discrete, compositional, and symbolic representations through attractor dynamics.** Andrew Nam, Eric Elmoznino, Nikolay Malkin, Chen Sun, Yoshua Bengio, Guillaume Lajoie *NeurIPS Workshop oral*
- 2023 **Consciousness in Artificial Intelligence: Insights from the Science of Consciousness.** Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, Rufin VanRullen *Preprint, condensed 2025 version published in Trends in Cognitive Sciences*
- 2023 **Scene context is predictive of unconstrained object similarity judgments.** Caterina Magri, Eric Elmoznino, Michael F. Bonner *Cognition*
- 2023 **Learning Macro Variables with Auto-encoders.** Maitreyi Swaroop, Eric Elmoznino, Dhanya Sridhar *Neurips Workshop poster*
- 2020 **Visual representations derived from multiplicative interactions.** Eric Elmoznino & Michael F. Bonner *NeurIPS Workshop poster*
- 2019 **A new procedure, free from human assessment that automatically grades some facial skin structural signs.** Jiang R., Kezele I., Levinshstein A., Flament F., Zhang J., Elmoznino E., Ma J., Ma J., Coquide J., Arcin V., Omoyuri E., Aarabi P. *International Journal of Cosmetic Science*

Invited Talks & Podcasts

2026	Compression & intelligence — <i>Center for Neuroscience Imaging Research, South Korea</i>
2026	Compositional languages — <i>Cognition theory journal club, NYU</i>
2025	Compositional languages — <i>Analogy group</i>
2025	Compositional languages — <i>Jennifer Hu Lab, Johns Hopkins University</i>
2025	Good visual representations are high-dimensional — <i>Martin Hebart Lab, Max Planck Institute</i>
2025	Could AIs be Conscious? — <i>New Horizons podcast</i>
2024	Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability — <i>Models of Consciousness conference</i>
2024	Consciousness, ineffability, and AI safety — <i>AI Safety Reading Group, Mila</i>
2023	Sampling discrete objects through continuous attractor dynamics — <i>GFlowNet Reading Group, Mila</i>
2023	Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability — <i>Computational Phenomenology Group</i>
2023	Why can't we describe our conscious experiences? An attractor dynamics perspective of the ineffability of qualia — <i>University of Toronto guest lecture</i>
2023	Why can't we describe our conscious experiences? An information theoretic attractor dynamics perspective of ineffability — <i>Active Inference Institute podcast</i>
2020	How does the brain work? Cognitive science research — <i>SABES</i>
2020	Introduction to Programming with Python — <i>UofTHacks</i>

Supervision & Mentoring

2025-Now	Tejas Kasetty Master's (Computer Science)	Mila, Montreal
2025-Now	Yinhan Lu Master's (Computer Science)	Mila, Montreal
2023	Maitreyi Swaroop Master's (Mathematics and Computing)	Mila, Montreal
2021-22	Atlas Kazemian Master's (Cognitive Science)	Johns Hopkins University, Baltimore
2020-21	Adyant Balaji Undergraduate (Computer Engineering & Cognitive Science)	Johns Hopkins University, Baltimore
2019-20	Maro Maged Doce Undergraduate (Neuroscience)	Johns Hopkins University, Baltimore

Academic service

2026-Now	Mila Tea Talk Organizer Invite and host speakers at Mila "tea talks", the research institute's primary seminar series	Mila, Montreal
2026	Area Chair for InterpScience Coordinate reviewer-author discussions and provide final acceptance decision ○ NeurIPS workshop	
2026-Now	Reviewer for TMLR Reviewer for machine learning journal, at least 6 papers per year	
2022-Now	Reviewer for AI conferences Regular reviewer for NeurIPS, ICLR, and ICML	
2015-16	Finance Chair Elected by peers at the University of Toronto to manage the budget and plan social activities	Electrical and Computer Engineering Student Association, Toronto
2014-15	Class Representative Elected by peers at the University of Toronto to represent student interest at faculty meetings	Electrical and Computer Engineering Student Association, Toronto

Highlighted research directions

- 2023-Now **Compositionality, structure, and complexity**
Deriving a formal quantitative definition of compositionality using ideas from algorithmic information theory, compression, and minimum description length
- 2024-Now **Open-ended knowledge growth and curriculum construction**
Training a policy with an intrinsic reward derived from algorithmic information theory to optimally explore complex environments and seek new knowledge efficiently
- 2023-Now **Compositional attractor models of human thought**
Learning discrete and compositional models of conscious human thought using neural network attractor dynamics, from which symbolic representations can naturally emerge
- 2024-Now **In-context learning**
Developing normative theories of in-context learning, extracting in-context model representations, estimating the amount of structure an in-context model has learned from data
- 2024-Now **The hard problem of consciousness and information semantics**
Building formal theories of information semantics/meaning and using them to resolve aspects of the hard problem of consciousness (i.e., conscious experience)

Patents

- 2022 **System and method for image processing using deep neural networks.** Levinshtein A., Chang C., Phung E., Kezele I., Guo W., Elmoznino E., Jiang R., Aarabi P. *U.S. Patent No. 11216988*
- 2021 **Image-to-image translation using unpaired data for supervised learning.** Elmoznino E., Kezele I., Aarabi P. *U.S. Patent Application No. 17096774*
- 2020 **System and method for augmented reality using conditional cycle-consistent generative image-to-image translation models.** Elmoznino E., Ma H., Kezele I., Phung E., Levinshtein A., Aarabi P. *U.S. Patent Application No. 16683398*
- 2020 **Machine image colour extraction and machine image construction using an extracted colour.** Elmoznino E., Aarabi P., Zhang Y. *U.S. Patent Application No. 16854975*
- 2020 **Automatic image-based skin diagnostics using deep learning.** Jiang R., Ma J., Ma H., Elmoznino E., Kezele I., Levinshtein A., Charbit J., Despois J., Perrot M., Antoinin F., Flament R.S., Parham A. *U.S. Patent Application No. 16702895*